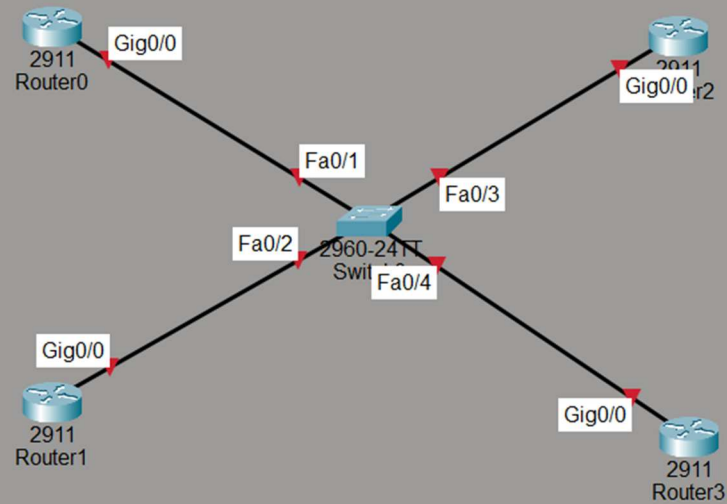




```

Router(config)#hostname IOU1
IOU1(config)#interface Loopback0

IOU1(config-if)# ip address 10.1.11.1 255.255.255.255
IOU1(config-if)#interface GigabitEthernet0/0
IOU1(config-if)# ip address 10.1.1.1 255.255.255.0
IOU1(config-if)# shutdown
IOU1(config-if)#router ospf 1
IOU1(config-router)# router-id 1.1.1.1
IOU1(config-router)# network 10.1.1.0 0.0.0.255 area 0
IOU1(config-router)# network 10.1.11.1 0.0.0.0 area 0
IOU1(config-router)#end
%LINK-3-UPDOWN: Interface Loopback0, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface Loopback0, changed state to up

IOU1#
%SYS-5-CONFIG_I: Configured from console by console

IOU1#
IOU1#

```

```

IOU1#
IOU1#show ip ospf neighbor

Neighbor ID      Pri   State           Dead Time   Address      Interface
4.4.4.4          1     FULL/DR         00:00:34   10.1.1.4     GigabitEthernet0/0
2.2.2.2          1     FULL/DROTHER    00:00:31   10.1.1.2     GigabitEthernet0/0
3.3.3.3          1     FULL/DROTHER    00:00:33   10.1.1.3     GigabitEthernet0/0
IOU1#show ip ospf interface GigabitEthernet0/0

```

```

GigabitEthernet0/0 is up, line protocol is up
Internet address is 10.1.1.1/24, Area 0
Process ID 1, Router ID 1.1.1.1, Network Type BROADCAST, Cost: 1
Transmit Delay is 1 sec, State BDR, Priority 1
Designated Router (ID) 4.4.4.4, Interface address 10.1.1.4
Backup Designated Router (ID) 1.1.1.1, Interface address 10.1.1.1
Timer intervals configured, Hello 10, Dead 40, Wait 40, Retransmit 5
Hello due in 00:00:07
Index 2/2, flood queue length 0
Next 0x0(0)/0x0(0)
Last flood scan length is 1, maximum is 1
Last flood scan time is 0 msec, maximum is 0 msec
Neighbor Count is 3, Adjacent neighbor count is 3
  Adjacent with neighbor 4.4.4.4 (Designated Router)
  Adjacent with neighbor 2.2.2.2
  Adjacent with neighbor 3.3.3.3
Suppress hello for 0 neighbor(s)
IOU1#

```

```

IOU4#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
IOU4(config)#interface GigabitEthernet0/0
IOU4(config-if)# no shutdown

IOU4(config-if)#end
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

IOU4#
%SYS-5-CONFIG_I: Configured from console by console

IOU4#

```

```

IOU1#show ip ospf neighbor

Neighbor ID      Pri   State           Dead Time   Address      Interface
2.2.2.2          1     FULL/DROTHER    00:00:30   10.1.1.2     GigabitEthernet0/0
3.3.3.3          1     FULL/BDR        00:00:32   10.1.1.3     GigabitEthernet0/0
IOU1#

```

```

IOU4#
IOU4#show ip ospf neighbor

Neighbor ID      Pri   State           Dead Time   Address      Interface
1.1.1.1          1     FULL/DR         00:00:32   10.1.1.1     GigabitEthernet0/0
2.2.2.2          1     2WAY/DROTHER    00:00:34   10.1.1.2     GigabitEthernet0/0
3.3.3.3          1     FULL/BDR        00:00:36   10.1.1.3     GigabitEthernet0/0
IOU4#
IOU4#

```

```

IOU1#
IOU1#show ip ospf interface GigabitEthernet0/0

GigabitEthernet0/0 is up, line protocol is up
 Internet address is 10.1.1.1/24, Area 0
 Process ID 1, Router ID 1.1.1.1, Network Type BROADCAST, Cost: 1
 Transmit Delay is 1 sec, State DR, Priority 200
 Designated Router (ID) 1.1.1.1, Interface address 10.1.1.1
 Backup Designated Router (ID) 4.4.4.4, Interface address 10.1.1.4
 Timer intervals configured, Hello 10, Dead 40, Wait 40, Retransmit 5
   Hello due in 00:00:01
 Index 2/2, flood queue length 0
 Next 0x0(0)/0x0(0)
 Last flood scan length is 1, maximum is 1
 Last flood scan time is 0 msec, maximum is 0 msec
 Neighbor Count is 3, Adjacent neighbor count is 3
   Adjacent with neighbor 4.4.4.4 (Backup Designated Router)
   Adjacent with neighbor 2.2.2.2
   Adjacent with neighbor 3.3.3.3
 Suppress hello for 0 neighbor(s)
IOU1#
IOU1#

```

```

IOU1#show ip ospf
Routing Process "ospf 1" with ID 5.5.5.5
Supports only single TOS(TOS0) routes
Supports opaque LSA
SPF schedule delay 5 secs, Hold time between two SPFs 10 secs
Minimum LSA interval 5 secs. Minimum LSA arrival 1 secs
Number of external LSA 0. Checksum Sum 0x000000
Number of opaque AS LSA 0. Checksum Sum 0x000000
Number of DCbitless external and opaque AS LSA 0
Number of DoNotAge external and opaque AS LSA 0
Number of areas in this router is 1. 1 normal 0 stub 0 nssa
External flood list length 0
  Area BACKBONE(0)
    Number of interfaces in this area is 2
    Area has no authentication
    SPF algorithm executed 53 times
    Area ranges are
      Number of LSA 6. Checksum Sum 0x0285c7
      Number of opaque link LSA 0. Checksum Sum 0x000000
      Number of DCbitless LSA 0
      Number of indication LSA 0
      Number of DoNotAge LSA 0
      Flood list length 0

```

```

IOU1#

```

```

IOU1#show ip ospf interface brief

```

Interface	PID	Area	IP Address/Mask	Cost	State	Nbrs	F/C
Lo0	1	0	10.1.11.1/255.255.255.255	1	WAIT	0/0	
Gig0/0	1	0	10.1.1.1/255.255.255.0	1	DR	0/0	

```

IOU1#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

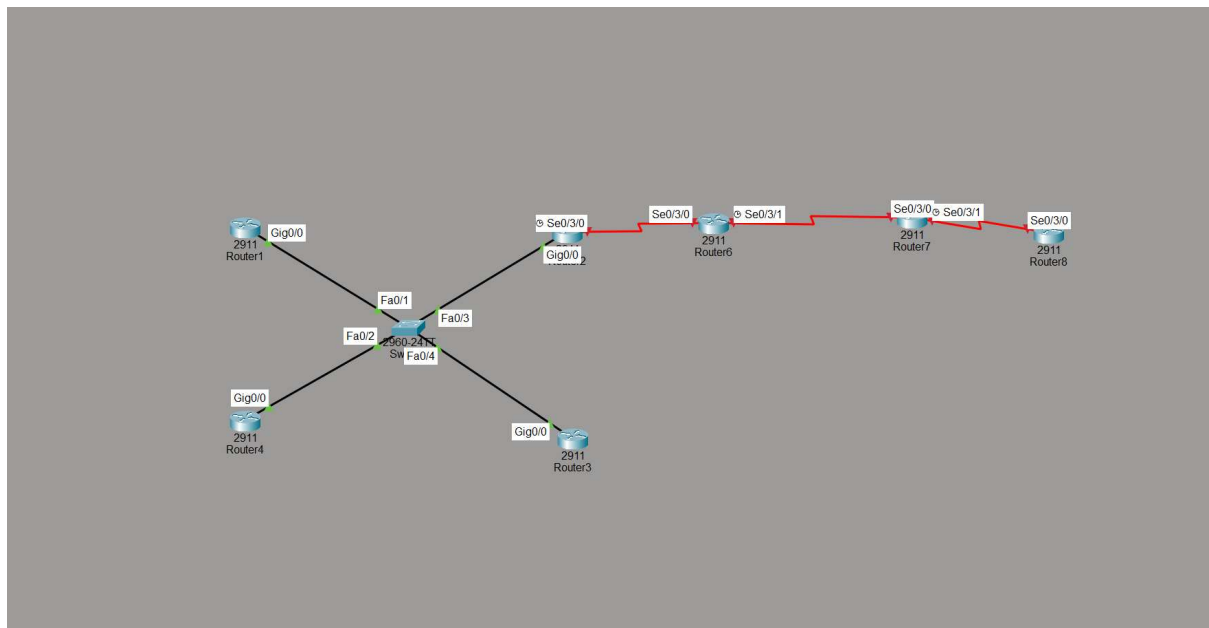
Gateway of last resort is not set

10.0.0.0/8 is variably subnetted, 6 subnets, 2 masks
C    10.1.1.0/24 is directly connected, GigabitEthernet0/0
L    10.1.1.1/32 is directly connected, GigabitEthernet0/0
C    10.1.11.1/32 is directly connected, Loopback0
O    10.1.22.1/32 [110/2] via 10.1.1.2, 00:13:19, GigabitEthernet0/0
O    10.1.33.1/32 [110/2] via 10.1.1.3, 00:02:26, GigabitEthernet0/0
O    10.1.44.1/32 [110/2] via 10.1.1.4, 00:02:26, GigabitEthernet0/0

IOU1#show ip route ospf
10.0.0.0/8 is variably subnetted, 6 subnets, 2 masks
O    10.1.22.1 [110/2] via 10.1.1.2, 00:13:21, GigabitEthernet0/0
O    10.1.33.1 [110/2] via 10.1.1.3, 00:02:28, GigabitEthernet0/0
O    10.1.44.1 [110/2] via 10.1.1.4, 00:02:28, GigabitEthernet0/0

IOU1#

```



```

IOU6#show ip ospf neighbor

```

Neighbor ID	Pri	State	Dead Time	Address	Interface
2.2.2.2	0	FULL/ -	00:00:38	10.99.1.1	Serial0/3/0
7.7.7.7	0	FULL/ -	00:00:37	10.99.2.2	Serial0/3/1

```

IOU6#

```

```

IOU6#show ip ospf interface Serial0/3/0

Serial0/3/0 is up, line protocol is up
 Internet address is 10.99.1.2/30, Area 0
 Process ID 1, Router ID 6.6.6.6, Network Type POINT-TO-POINT, Cost: 64
 Transmit Delay is 1 sec, State POINT-TO-POINT,
 Timer intervals configured, Hello 10, Dead 40, Wait 40, Retransmit 5
   Hello due in 00:00:02
 Index 2/2, flood queue length 0
 Next 0x0(0)/0x0(0)
 Last flood scan length is 1, maximum is 1
 Last flood scan time is 0 msec, maximum is 0 msec
 Neighbor Count is 1 , Adjacent neighbor count is 1
   Adjacent with neighbor 2.2.2.2
 Suppress hello for 0 neighbor(s)
IOU6#

```

```

IOU1#
IOU1#show ip route ospf
 10.0.0.0/8 is variably subnetted, 10 subnets, 3 masks
O       10.1.22.1 [110/2] via 10.1.1.2, 00:05:11, GigabitEthernet0/0
O       10.1.33.1 [110/2] via 10.1.1.3, 00:05:11, GigabitEthernet0/0
O       10.1.44.1 [110/2] via 10.1.1.4, 00:04:18, GigabitEthernet0/0
O       10.1.66.1 [110/66] via 10.1.1.2, 00:03:12, GigabitEthernet0/0
O       10.1.77.1 [110/130] via 10.1.1.2, 00:03:02, GigabitEthernet0/0
O       10.99.1.0 [110/65] via 10.1.1.2, 00:03:22, GigabitEthernet0/0
O       10.99.2.0 [110/129] via 10.1.1.2, 00:03:12, GigabitEthernet0/0
O*E2 0.0.0.0/0 [110/1] via 10.1.1.2, 00:02:52, GigabitEthernet0/0
IOU1#

```

```

IOU6#show ip ospf neighbor

Neighbor ID      Pri   State           Dead Time   Address        Interface
7.7.7.7          0    FULL/ -         00:00:30    10.99.2.2      Serial0/3/1
IOU6#

```

OSPF ERWEITERTE TOPOLOGIE - KONFIGURATIONSSKRIPT

ROUTER 1 (IOU1)

enable

configure terminal

hostname IOU1

interface Loopback0

ip address 10.1.11.1 255.255.255.255

```
no shutdown
```

```
interface GigabitEthernet0/0
```

```
ip address 10.1.1.1 255.255.255.0
```

```
ip ospf priority 1
```

```
no shutdown
```

```
router ospf 1
```

```
router-id 1.1.1.1
```

```
network 10.1.1.0 0.0.0.255 area 0
```

```
network 10.1.11.1 0.0.0.0 area 0
```

```
end
```

```
copy running-config startup-config
```

ROUTER 2 (IOU2)

```
enable
```

```
configure terminal
```

```
hostname IOU2
```

```
interface Loopback0
```

```
ip address 10.1.22.1 255.255.255.255
```

```
no shutdown
```

```
interface GigabitEthernet0/0
```

```
ip address 10.1.1.2 255.255.255.0
```

```
ip ospf priority 1
```

```
no shutdown
```

```
interface Serial0/3/0
```

```
ip address 10.99.1.1 255.255.255.252
```

```
clock rate 64000
```

```
no shutdown
```

```
router ospf 1
```

```
router-id 2.2.2.2
```

```
network 10.1.1.0 0.0.0.255 area 0
```

```
network 10.99.1.0 0.0.0.3 area 0
```

```
network 10.1.22.1 0.0.0.0 area 0
```

```
end
```

```
copy running-config startup-config
```

ROUTER 3 (IOU3)

```
enable
```

```
configure terminal
```

```
hostname IOU3
```

```
interface Loopback0
```

```
ip address 10.1.33.1 255.255.255.255
```

```
no shutdown
```

```
interface GigabitEthernet0/0  
ip address 10.1.1.3 255.255.255.0  
ip ospf priority 1  
no shutdown
```

```
router ospf 1  
router-id 3.3.3.3  
network 10.1.1.0 0.0.0.255 area 0  
network 10.1.33.1 0.0.0.0 area 0  
end  
copy running-config startup-config
```

ROUTER 4 (IOU4)

```
enable  
configure terminal  
hostname IOU4
```

```
interface Loopback0  
ip address 10.1.44.1 255.255.255.255  
no shutdown
```

```
interface GigabitEthernet0/0  
ip address 10.1.1.4 255.255.255.0  
ip ospf priority 1  
no shutdown
```

```
router ospf 1
router-id 4.4.4.4
network 10.1.1.0 0.0.0.255 area 0
network 10.1.44.1 0.0.0.0 area 0
end
copy running-config startup-config
```

ROUTER 6 (IOU6)

```
enable
configure terminal
hostname IOU6

interface Loopback0
ip address 10.1.66.1 255.255.255.255
no shutdown

interface Serial0/3/0
ip address 10.99.1.2 255.255.255.252
no shutdown

interface Serial0/3/1
ip address 10.99.2.1 255.255.255.252
clock rate 64000
no shutdown
```

```
router ospf 1
router-id 6.6.6.6
network 10.99.1.0 0.0.0.3 area 0
network 10.99.2.0 0.0.0.3 area 0
network 10.1.66.1 0.0.0.0 area 0
end
copy running-config startup-config
```

ROUTER 7 (IOU7)

```
enable
configure terminal
hostname IOU7

interface Loopback0
ip address 10.1.77.1 255.255.255.255
no shutdown

interface Serial0/3/0
ip address 10.99.2.2 255.255.255.252
no shutdown

interface Serial0/3/1
ip address 10.99.3.1 255.255.255.252
no shutdown
```

```
ip route 0.0.0.0 0.0.0.0 Serial0/3/1
```

```
router ospf 1
```

```
router-id 7.7.7.7
```

```
default-information originate
```

```
network 10.99.2.0 0.0.0.3 area 0
```

```
network 10.1.77.1 0.0.0.0 area 0
```

```
end
```

```
copy running-config startup-config
```

ROUTER 8 (IOU8)

```
enable
```

```
configure terminal
```

```
hostname IOU8
```

```
interface Loopback0
```

```
ip address 10.1.88.1 255.255.255.255
```

```
no shutdown
```

```
interface Serial0/3/0
```

```
ip address 10.99.3.2 255.255.255.252
```

```
no shutdown
```

```
end
```

```
copy running-config startup-config
```